TRISTAN COIGNION

Sustainable AI & Software Engineering researcher

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Research Interests

My research sits at the intersection of **software engineering** and **environmental sustainability**, with a focus on understanding and reducing the ecological footprint of modern AI systems.

Green AI & Energy Consumption Measurement. I study the energy consumption of AI systems across their full lifecycle, from training to inference, with a particular interest in developing rigorous, reproducible measurement methodologies. This includes designing empirical frameworks to capture the energy cost of AI-assisted developer tools (e.g., code assistants) during real development sessions, and proposing standardized models for reporting AI-related energy metrics.

Large Language Models for Code. I investigate the capabilities and costs of LLMs applied to software engineering tasks, including automated code generation, performance optimization, and code quality assessment. I am interested in the trade-off between the utility that LLMs bring to developers and the environmental cost they incur, both during use and through the code they produce.

Environmental Impact of Generative AI Media. My current work extends these questions to generative AI for video, exploring how to estimate and model the environmental impacts of AI video generation systems, accounting for the complexity of media-specific workloads and usage patterns.

Education

2022–2025 **PhD, Computer Science & Engineering**, *Inria and University of Lille*, Lille.
Large Language Models for code, Software sustainability, Energy consumption, Green AI

2020–2022 **Master of Software Engineering**, *University of Lille*, highest honours.

2017–2020 **Bachelor of Computer Science**, *University of Lille*, highest honours.

2014–2017 **Scientific Baccalaureate**, *Institut de Genech*, highest honours.

Publications

In Conference Proceedings

2025 Tristan Coignion, Clément Quinton, and Romain Rouvoy. When faster isn't greener: The hidden costs of llm-based code optimization. In *ASE 2025 (Core Rank A*) (to be published)*, 2025.

2024 Tristan Coignion, Clément Quinton, and Romain Rouvoy. A Performance Study of LLM-Generated Code on Leetcode. In *EASE 2024 (Core Rank A), Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering*, pages 79–89, Salerno Italy, June 2024. ACM. Available at `https://dl.acm.org/doi/10.1145/3661167.3661221`.

Prepublications

2024 Tristan Coignion, Clément Quinton, and Romain Rouvoy. Green My LLM: Studying the key factors affecting the energy consumption of code assistants, November 2024. Available at `https://arxiv.org/pdf/2407.21579`. Under review.

Experience

LaBRI, University of Bordeaux

2026 **Post-doc: *Estimating and modeling the environmental impacts of AI video generation***.

Supervisor **Prof. Aurélie Bugeau**, *Full Professor, University of Bordeaux* (Personal Webpage)

Supervisor **Prof. Anne-Laure Ligozat**, *Full Professor, LISN and ensIIE* (Personal Webpage)

Inria, University of Lille

October 2022 – 2025 ***Studying the energy consumption of Large Language Models for code***.
- We studied the performance of code generated by multiple Large Language Models.
- We studied the energy cost of using a code assistant like GitHub Copilot with human participants. We proposed a dataset of development traces, as well as a novel methodology to evaluate the energy consumption of code assistants.
- We studied the optimization of code using LLMs as well as the cost of the optimization.

Advisor **Prof. Romain Rouvoy**, *Full Professor, University of Lille* (*Personal Webpage*)

Advisor **Prof. Clément Quinton**, *Associate Professor, University of Lille* (*Personal Webpage*)

October 2022 – 2025 **DISTILLER ANR Project contribution**.
Contributed to the DISTILLER ANR project. My work focused on recommender systems for more sustainable software artifacts.

Boavizta

December 2024 – present **Active member of the association**.
- I contributed to the BoAmps project. This project aims at providing a common model for reporting energy consumption metrics related to AI usage.

Others

2020 – 2022 **Software engineer**, *Glaz Tech+Fi (defunct)*, Lille.
Software development on the Salesforce Platform for finance users. Team and project management.

Posters and presentations

2025 **Poster - Green My LLM: Studying the key factors affecting the energy consumption of code assistants**, *Green Days 2025*, Rennes, [*Poster*].

2025 **Green My LLM: How much does your copilot eat?**, *GT-GLIA 2025*, Rennes, [*Slides*].

2024 **A performance Study of LLM-Generated Code on Leetcode**, *EASE 2024 (Core Rank A)*, Salerno, [*Slides*].

2024 **A performance Study of LLM-Generated Code on Leetcode**, *Green Days 2024*, Toulouse, [*Slides*].

Computer skills

Programming Languages: Python (Pandas/Polars, Matplotlib, Scikit-learn), Java, C, C++

Web Technologies: HTML 5, Javascript, CSS, React

Services

Teaching at the University of Lille

Spring 2025 **Distributed systems (27h)**, Microservice architecture, software resilience, fault tolerance, etc..

Spring 2024, 2025 **Operating system architecture (36h)**, Understanding how a kernel works and implementing a process scheduler from scratch in C.

Fall 2024 **Object oriented conception (18h)**, Java programming, design pattern, software design.

Fall 2023: **Introduction to web development (31.5h)**, HTML5, CSS, JS.

Reviewing

2025 **Software: Practice and Experience**.

2025 **IEEE/ACM International Conference on Software Engineering**.

2024 **ACM Transactions on Software Engineering and Methodology**.

References

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